Pascal's triangle

1 The 4th row of Pascal's triangle consists of the numbers $1, 4, 6, 4, 1$.

a Write down the numbers in the 5th row of Pascal's triangle.

b Find the sum of the entries in the 5th row of Pascal's triangle.

c Rewrite each element of the 5th row of Pascal's triangle as an ${}_nC_r$ coefficient.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1,\ 8,\ \square,\ 56,\ 70,\ \square,\ 28,\ \square,\ 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the distribution of $(x+y)^7$?

4 State the number of terms in the following distributions:

a $(4x+y)^6$

b $(m+y)^8$

5 Consider the distribution of $(y+k)^4$.

a How many terms will there be in the distribution?

b Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following distributions:

a $(1+x)^4 = \square + \square x + \square x^2 + \square x^3 + \square x^4$

b $(x+m)^4 = \square x^4 m^0 + \square x^3 m^1 + \square x^2 m^2 + \square x^1 m^3 + \square x^0 m^4$

c $(5+b)^5 = \square \times 5^5 b^0 + \square \times 5^4 b^1 + \square \times 5^3 b^2 + \square \times 5^2 b^3 + \square \times 5^1 b^4 + \square \times 5^0 b^5$

d $(ka-b)^6 = \square (ka)^4 (-b)^0 + \square (ka)^3 (-b)^1 + \square (ka)^2 (-b)^2 + \square (ka)^1 (-b)^3 + \square (ka)^0 (-b)^4 + \square (ka)^0 (-b)^5 + \square (ka)^0 (-b)^6$

7 Distribute the following using Pascal's triangle:

a $(p+q)^5$

b $(c-d)^8$

c $(y+3)^4$

d $(u+4v)^6$

e $\left(1+\dfrac{4}{y}\right)^3$

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8 Evaluate the following:

a $\binom{6}{5}$ b $\binom{14}{1}$ c $\binom{10}{9}$ d $\binom{8}{0}$

Binomial theorem

9 Consider the distribution of $(a-4)^5$. Will the 2nd term be positive or negative?

10 Complete the following distribution:

$$(4p+3q)^3 = {}_3C_0\,(4p)^{\square}\,(3q)^0 + {}_3C_1\,(4p)^{\square}\,(3q)^1 + {}_3C_2\,(4p)^{\square}\,(3q)^2 + {}_3C_3\,(4p)^{\square}\,(3q)^3$$

11 Consider the binomial $(4x+3y)^4$.

a State the first term in the distribution.

b State the last term in the distribution.

12 Which two terms in the distribution of $(u+v)^{11}$ have a coefficient of ${}_{11}C_9$?

13 Find the specified term for the following distributions:

a Fifth term of $(b+2)^7$.

b Fifth term of $(c+d)^9$.

c Fifth term of $(3u-v)^6$.

d Tenth term of $\left(u+\frac{1}{2}\right)^{12}$.

e Seventh term of $\left(\frac{3y}{2}-\frac{2}{3y}\right)^{10}$.

f Eleventh term of $\left(2x+y^2\right)^{13}$.

g Sixteenth term of $\left(x-y^5\right)^{19}$.

h Middle term of $\left(4x^2+3y^3\right)^8$.

i Middle term of $\left(-3x^{-3}+2y^{-2}\right)^4$.

j Constant term of $\left(2y-\frac{1}{y^3}\right)^8$.

14 Use the binomial theorem to distribute the following expressions:

a $(1+x)^4$ b $(2+b)^5$ c $(p+q)^5$ d $(c-d)^8$

e $(u+4v)^6$ f $(y+3)^4$ g $(ka-b)^6$ h $(y-5)^3$

i $\left(y-\frac{1}{3}\right)^3$ j $\left(2+\frac{y}{2}\right)^3$ k $(3y+2r)^3$ l $\left(a-\frac{1}{a}\right)^3$

m $\left(u^2+3v^2\right)^3$ n $\left(1+\frac{4}{y}\right)^3$

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15 The first three terms of $(x+y)^{10}$ are x^{10}, 10[redacted]nd $45x^8y^2$. Using the symmetry of the coefficients, write the last three terms of the distribution.

16 In the distribution of $(m+k)^n$, the coefficient of the 9th term is 45.

a Find the value of n.

b Find the 9th term.

17 If the eighth and tenth terms in the distribution of $(x+y)^n$ have the same coefficients, find the value of n.

18 Find the coefficient of a^3b^6 in the distribution of $\left(2a-\frac{b}{2}\right)^9$.

19 Consider the distribution of $(1+x)^9$. State the coefficient of the 4th term in the form ${}_nC_r$.

20 Find:

a The term in $(3+\sqrt{x})^{10}$ that contains x^4.

b The term in $(c+d)^{10}$ that contains c^3.

c The term in $(u+v)^{11}$ that contains v^8.

d The term in $(u+v)^{15}$ that contains u^5.

21 Find the coefficient of x^{17} in the distribution of $\left(3x^2+2x\right)^{11}$.

22 Find the value of n such that the distribution of $(a+b)^n$ contains the term $84a^6b^3$.

23 A particular term in the distribution of $\left(3a^2+\frac{p}{b}\right)^4$ is $\frac{96a^2}{b^3}$, for some constant p. Find the value of p.

24 By considering $(x+y)^4=(x+y)(x+y)(x+y)(x+y)$, in how many ways can three factors of x and one factor of y be chosen from the distribution to form the term which contains x^3y^1? Write your answer in the form ${}_nC_r$.

25 Consider the distribution of $(a+b)^5$.

a State the coefficient of the 3rd term in the form ${}_nC_r$.

b Evaluate the coefficient of the 3rd term.

c By considering the symmetry of ${}_nC_r$, which other term has the same coefficient as the 3rd term?

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26 Complete the following distribution by dete[redacted]ng the missing power and binomial coefficient of the $(k+1)$th term:

$$(3-x)^8 = {}_8C_0 \times 3^8 (-x)^0 + {}_8C_1 \times 3^7 (-x)^1 + \ldots + \square \times 3^{8-k} (-x)^{\square} + \ldots + {}_8C_8 \times 3^0 (-x)^8$$

27 Consider the binomial series:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

a Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.

b Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.

c Use the first five terms to approximate $\dfrac{1}{(1.06)^2}$ to the nearest thousandth.

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